Run report: run-20261023T0121Z (rho.negation against rho.negation_lanes; external rho)

A descriptive record of the run. The results report is `../../speedups-20261023/README.md`.

What ran

item	value
spec	`../../speedups-20261023/rho-lanes/spec.json` (spec `ECS1h17511a88c065`): every 16th curve model of `run/specs/spec.json` (48: both sides, every level), 4 `planted_shared` targets each
arms	`rho-neg` = `rho.negation` (reference); `rho-neg-lanes` = `rho.negation_lanes`, `lanes = 32` (candidate)
measurement	warmup 1, rounds 3, order shuffle, seed 20261022, timeout 600 s
run	`ecbench run --cpus auto --allow-busy`; session `ECBS1h45f937d2f95a`, 01:21:26–01:29:02 UTC, 2026-10-23
host	Intel Xeon Processor @ 2.10 GHz, 4 vCPU shared cloud VM, Linux; isolation earned L1 / L2 per run
ecbench binary	sha256 `d9a6ca939982404c…`; sessions' git `bec9df9e8`, dirty (the lanes walk, committed as `d81da47d5`)
external	`rho-lanes/external_ca.sh`: `ca solve --alg rho --threads 1` (`aburan28/cryptanalysis` 1aa48b5, `ca` 0.1.0), `taskset -c 2`, seeds 1–3, on the session's 192 workload targets; run after the session (binary sha256 in `rho-lanes/external_ca.binary.sha256`, recorded afterwards)

Runs

arm	executions	measured (non-warmup)	verified	failures
`rho-neg`	768	576	768	0
`rho-neg-lanes`	768	576	768	0
external `ca`	576	576	576 recovered the planted scalar	0

The audit is `ecbench verify --replay 24`: `audit OK: 1536 records, 1536 verified, 24 replays reproduced, 0 problems` (`rho-lanes/session.audit.json`).

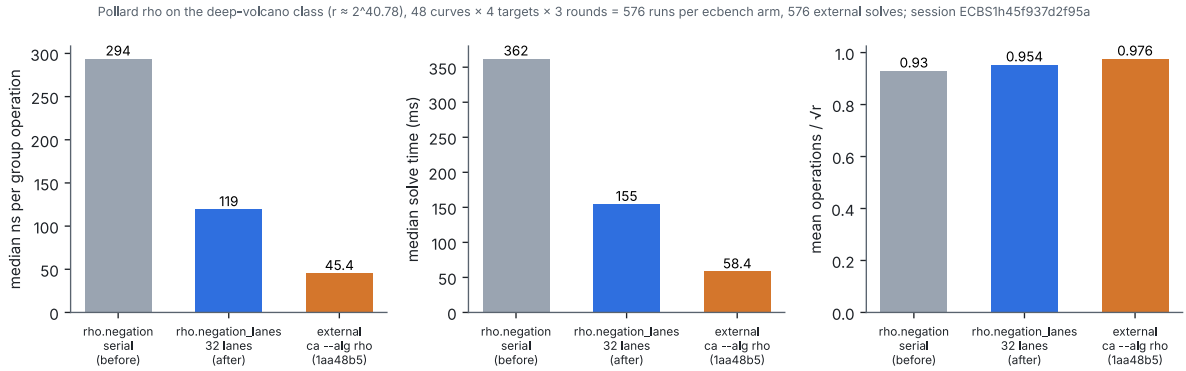
Recorded values (from `summary.json`)

quantity	`rho-neg`	`rho-neg-lanes`	external `ca`
median solve wall time	362.1 ms	154.6 ms	58.4 ms (its own timer)
median wall time per counted operation	293.9 ns	119.4 ns	45.4 ns (its own count)
mean S (GAE / √r)	0.9303	0.9538	0.976 (ops / √r)

The run is paired by `ecbench compare`:

- counted operations, lanes over serial: 1.0252 [1.0213, 1.0295], cluster interval over 576 pairs;
- median wall ratio: 0.4193 [0.4173, 0.4214], over all 576 pairs (306 of them at L2 on both sides);
- measurement quality: 58/100.

The §5a band is in `rho-lanes/band.json`.



Raw records

- `../../speedups-20261023/rho-lanes/session/records.jsonl.gz` (pinned by `../../speedups-20261023/MANIFEST.sha256`), plus `session.json`, `host.json`, `plan.json.gz`
- `../../speedups-20261023/rho-lanes/external_ca.jsonl`, `external_ca.inputs.tsv`
- `../../speedups-20261023/rho-lanes/session/comparisons/rho-neg__rho-neg-lanes.json`, `band.json`